

Autocorrelation and Mean Reversion

Serial dependence, mean reversion evidence, decay, validation, and regime-break controls.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

Table of contents

INTERMEDIATE

ORIENTATION

Title	1
How to use this manual	5
Notation and dependence contract	6
Dependence-to-trade research workflow	7

SERIAL DEPENDENCE FOUNDATIONS

Serial dependence is conditional structure	8
Autocovariance and lag geometry	9
Autocorrelation function	10
Sample ACF and finite-sample uncertainty	11
Partial autocorrelation	12
Positive and negative autocorrelation	13
Persistence is not mean reversion	14
Mechanical autocorrelation	15

Table of contents

INTERMEDIATE

MEAN REVERSION MODELS

Mean reversion needs a defensible anchor	16
AR(1) as a discrete reversion model	17
Half-life and decay speed	18
Ornstein-Uhlenbeck process	19
Z-scores and standardized deviations	20
Crossings and first-passage time	21
Cointegrated spreads	22
Error-correction dynamics	23
Nonlinear, asymmetric, and regime reversion	24

ESTIMATION AND VALIDATION

Window length is a bias-variance decision	25
Persistence parameter uncertainty	26
Stationarity before reversion	27
Residual whiteness and Ljung-Box tests	28
Variance ratio as a horizon diagnostic	29
Hurst exponent and long-memory claims	30
Lag and threshold multiplicity	31
Walk-forward validation of reversion	32

Table of contents

INTERMEDIATE

SYSTEM AND STRATEGY DESIGN

Point-in-time dependence features	33
Transaction costs define the tradable band	34
Signal-to-position mapping	35
Break risk and stop design	36
Portfolio aggregation of reversion trades	37
Production monitoring for reversion models	38
Governance and reproducible lineage	39

VISUAL LABS AND REVIEW

ACF signatures and competing stories	40
The full mean-reversion trade cycle	41
Choosing and interpreting a lag	42
Half-life sensitivity near the unit boundary	43
Deviation, costs, and the no-trade band	44
From market event to legal reversion fill	45
Method selection by research question	46
Failure modes and diagnostic actions	47
Formula sheet and interpretation map	48
Worked calculation: deviation to governed trade	49
Validation scorecard for a reversion strategy	50
Final active recall and system index	51

How to use this manual

INTERMEDIATE

1. NAME THE MEASURED OBJECT

Begin with the exact level, return, residual, volatility transform, or event count. Record grid and price convention because dependence can reverse when sampling changes.

2. SEPARATE DESCRIPTION FROM MECHANISM

Use ACF, PACF, variance ratios, and passage times to describe evidence. Challenge each story with overlap, stale quotes, bounce, breaks, and regime mixing.

3. TRANSLATE COEFFICIENTS INTO OPERATIONS

Convert ϕ or κ into holding-period ranges, capital duration, stops, and cost budgets. Carry parameter uncertainty through each translation.

4. PRESERVE CAUSAL TIMING

Fit centers, scales, hedge ratios, and lag models at each historical origin. The earliest legal action follows availability and computation.

5. VALIDATE THE ENTIRE TRADE

Evaluate entry, path, censoring, exit, slippage, inventory age, and break loss. A significant lag is only the beginning of the decision case.

6. CONNECT EVIDENCE TO CONTROLS

Define warning, reduced-risk, and stop responses for stale data, center drift, slow passage, cost deterioration, and common portfolio stress.

DECISION STANDARD

Promote only when anchor, decay, timing, costs, failure scenarios, portfolio interactions, monitors, fallback, and lineage are reproducible.

Notation and dependence contract

INTERMEDIATE

SYMBOL	OBJECT	MEANING
X_t	observed process	Level, return, spread, residual, or transformed state
μ_t	conditional center	Point-in-time equilibrium or baseline
γ_k	autocovariance	Lag-k covariance in the original squared units
ρ_k	autocorrelation	Standardized lag-k linear association
ϕ	AR persistence	Fraction of deviation expected to survive one period
κ	OU speed	Continuous-time restoring-force parameter
h_{half}	half-life	Expected time for a deviation to shrink by half
e_t	equilibrium residual	Observed value minus causal center
z_t	standardized residual	Deviation measured in causal scale units
τ_b	first-passage time	Time to reach a boundary or exit

LAG AND CLOCK DISCIPLINE

Lag one means one observation interval on the declared grid. Five-minute bars, business days, sessions, and event time represent different horizons even when the index is identical.

MINIMUM MODEL METADATA

Store universe, price convention, calendar, transform, availability, center, scale, lag set, fit cutoff, uncertainty, cost model, thresholds, earliest action, fallback, and owner.

Dependence-to-trade research workflow

INTERMEDIATE

1

Define the decision

Target, horizon, execution, loss, and capacity

2

Build the clock

Calendar, availability, gaps, and resampling

3

Create the residual

Anchor, hedge relation, center, and scale

4

Map dependence

ACF, PACF, variance ratio, and pair counts

5

Challenge mechanics

Overlap, bounce, staleness, and breaks

6

Estimate decay

AR or OU speed, uncertainty, and passage

7

Simulate the trade

Legal entry, costs, exits, stops, censoring

8

Walk forward

Refit every learned artifact by origin

9

Stress the portfolio

Common anchors, crowding, and liquidity

10

Govern production

Health tiers, fallback, rollback, and lineage

Serial dependence is conditional structure

INTERMEDIATE

ONE-SENTENCE MEANING

Serial dependence means later outcomes change distribution when earlier observations are known. It may appear in the mean, variance, tails, or event rate.

MEMORY JOGGER

Time order can carry information that an IID model discards. Name the variable, sampling grid, lag, and conditional feature before making a claim.

WHY IT MATTERS IN QUANT RESEARCH

Forecasts, confidence intervals, and validation depend on how much fresh information each row adds. Ignoring dependence can create false precision even when average forecasts are unbiased.

FORMULA INTUITION

Independence implies $\text{Cov}(X_t, X_{t-k}) = 0$ when moments exist. Zero covariance does not imply independence because nonlinear structure can remain.

SIMPLE MARKET EXAMPLE

Daily returns may have little mean autocorrelation while squared returns cluster strongly. Direction and volatility models therefore see different memory in one feed.

PYTHON / SYSTEM IMPLEMENTATION

Create separate diagnostics for returns, absolute returns, squared returns, signs, and counts. Version the observation grid because five-minute and daily structure answer different questions.

CONFUSIONS TO AVOID

Dependence is not automatically profitable predictability. A smooth rolling statistic can be caused by overlapping windows rather than the market.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data contract records event time, availability, sampling, and transformation. The registry links each claim to the exact series and decision horizon.

RELATED CONCEPTS AND QUICK RECALL

Related: IID assumption, conditional distribution, volatility clustering, information set. Recall task: Name two forms of dependence a return ACF misses. State which uncertainty estimate each one invalidates.

Autocovariance and lag geometry

INTERMEDIATE

ONE-SENTENCE MEANING

Autocovariance measures how deviations from the mean move together across a stated lag. Its magnitude retains the squared units and scale of the series.

MEMORY JOGGER

Pair a series with a delayed copy and measure joint movement. A lag is an interval on a declared clock, not merely an integer index.

WHY IT MATTERS IN QUANT RESEARCH

Autocovariance enters linear forecasts, long-run variance, and effective sample size. Scale changes can alter it even when standardized dependence is stable.

FORMULA INTUITION

$\gamma_k = \text{Cov}(X_t, X_{t-k}) = E[(X_t - \mu)(X_{t-k} - \mu)]$. Under weak stationarity it depends on lag k rather than calendar time.

SIMPLE MARKET EXAMPLE

A one-minute spread has positive lag-one covariance during quiet quoting but negative covariance after aggressive trades. The full-day average may describe neither mechanism.

PYTHON / SYSTEM IMPLEMENTATION

Estimate pair counts and covariance by rolling origin and session phase. Store units, demeaning, denominator, missing-pair policy, and legal availability.

CONFUSIONS TO AVOID

Large autocovariance may reflect large variance rather than strong standardized association. A changing center can also turn one estimate into an unstable mixture.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dependence service publishes covariance with variance, pair count, and state metadata. Consumers can distinguish a scale shock from changed lag behavior.

RELATED CONCEPTS AND QUICK RECALL

Related: autocorrelation, weak stationarity, long-run variance, lag operator. Recall task: Explain how doubled volatility changes autocovariance when autocorrelation is fixed. Identify the risk calculation that must change.

Autocorrelation function

INTERMEDIATE

ONE-SENTENCE MEANING

The autocorrelation function standardizes lag covariance and maps linear dependence across lags. Each point summarizes one delayed pairing under a stated sample.

MEMORY JOGGER

Read the ACF as a dependence fingerprint rather than isolated significance tests. Shape, decay, sign changes, and economic timing matter together.

WHY IT MATTERS IN QUANT RESEARCH

The ACF helps diagnose persistence, reversal, seasonality, and residual structure. It also informs how much less evidence a dependent sample contains than its row count.

FORMULA INTUITION

$\rho_k = \gamma_k / \gamma_0$ and lies between -1 and 1 when defined. The sequence must satisfy covariance-matrix constraints, so lag values are not freely chosen.

SIMPLE MARKET EXAMPLE

A spread ACF decaying from 0.70 to 0.50 to 0.35 suggests persistence. A return ACF negative only at lag one may instead reflect bid-ask bounce.

PYTHON / SYSTEM IMPLEMENTATION

Plot estimates with pair counts, bands, and marked market intervals. Compare raw, cleaned, and resampled versions so transformations do not masquerade as economics.

CONFUSIONS TO AVOID

An ACF is descriptive, not proof of stationarity or causality. Zero-centered bands are misleading after broad lag searches or under strong heteroskedasticity.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Exploration builds the fingerprint, while validation recomputes it inside training folds. Production monitors only predeclared material lags and shapes.

RELATED CONCEPTS AND QUICK RECALL

Related: correlogram, lag profile, persistence, spectral density. Recall task: Describe ACF shapes for white noise, a persistent AR process, and bid-ask bounce. State one ambiguity for each.

Sample ACF and finite-sample uncertainty

INTERMEDIATE

ONE-SENTENCE MEANING

The sample ACF is a noisy estimate from overlapping lagged pairs. Its uncertainty depends on dependence, tails, missing data, mean estimation, and lag selection.

MEMORY JOGGER

Every lag consumes observations and reuses many values. Later bars usually have fewer effective pairs and greater specification freedom.

WHY IT MATTERS IN QUANT RESEARCH

Small apparent correlations can be noise, while useful dependence can be missed in short histories. Decisions need intervals, stability, and effect size instead of one band crossing.

FORMULA INTUITION

$r_k = \frac{\sum_{t=k+1}^T (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^T (x_t - \bar{x})^2}$. Alternative denominators create documented small-sample differences.

SIMPLE MARKET EXAMPLE

With 250 observations, lag-one correlation of -0.11 may look notable under an IID shortcut. Heavy tails and clustering can widen the relevant uncertainty.

PYTHON / SYSTEM IMPLEMENTATION

Use block bootstrap or model simulation when IID bands are implausible. Report pair counts and rerun estimates over later, nonoverlapping periods.

CONFUSIONS TO AVOID

The common plus-or-minus $1.96/\sqrt{T}$ band is only a white-noise reference. It does not correct for selecting the best lag among many.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment artifact stores estimator, preprocessing, maximum lag, confidence method, and selection rule. This makes repeated research comparable.

RELATED CONCEPTS AND QUICK RECALL

Related: sampling variation, Bartlett bands, block bootstrap, effective observations. Recall task: Explain why lag twenty is less precise than lag one. Propose an uncertainty method for clustered daily returns.

Partial autocorrelation

INTERMEDIATE

ONE-SENTENCE MEANING

Partial autocorrelation measures the incremental linear association at lag k after controlling for intermediate lags. It asks whether that lag adds direct structure beyond shorter history.

MEMORY JOGGER

ACF contains direct and indirect paths, while PACF attempts to isolate the newest lag. Use both as design clues, not automatic order selectors.

WHY IT MATTERS IN QUANT RESEARCH

PACF can suggest autoregressive order and redundant lag features. In unstable markets it matters only when later forecast loss and coefficients also improve.

FORMULA INTUITION

PACF at k is the coefficient on $X_{(t-k)}$ in a regression of X_t on lags 1 through k . Estimation methods differ slightly in short samples.

SIMPLE MARKET EXAMPLE

An AR(2) can have an ACF decaying across many lags while PACF concentrates at lags one and two. That pattern suggests two direct linear memory terms.

PYTHON / SYSTEM IMPLEMENTATION

Compute PACF with a documented estimator and maximum lag. Fit candidate orders inside training windows and compare residual whiteness and later loss.

CONFUSIONS TO AVOID

A population cutoff is exact only for idealized stable models. Breaks, nonlinear effects, seasonality, and measurement noise distort the pattern.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Feature generation exposes a fixed lag set, while model selection chooses among approved structures. Metadata records the rule and training cutoff.

RELATED CONCEPTS AND QUICK RECALL

Related: AR order, Yule-Walker, lag regression, model selection. Recall task: Contrast lag-three ACF and PACF. Explain why neither alone justifies a third trading feature.

Positive and negative autocorrelation

INTERMEDIATE

ONE-SENTENCE MEANING

Positive autocorrelation means nearby deviations tend to share sign, while negative autocorrelation means they tend to alternate. Interpretation depends on the transformed series and horizon.

MEMORY JOGGER

Positive suggests continuation of the measured object and negative suggests reversal. Returns, spreads, volatility, and inventory can imply different trades from the same sign.

WHY IT MATTERS IN QUANT RESEARCH

Sign determines direction in linear forecasts and affects uncertainty of averages. Positive dependence reduces effective information, while negative dependence can stabilize averages.

FORMULA INTUITION

For stationary AR(1), $X_t = \phi X_{t-1} + \epsilon_t$ and $\rho_k = \phi^k$. Negative ϕ makes the correlation sign alternate with k .

SIMPLE MARKET EXAMPLE

Daily index returns may continue over months while transaction returns reverse at the tick level. Aggregation changes the dominant mechanism.

PYTHON / SYSTEM IMPLEMENTATION

Estimate by horizon, session, liquidity, and price convention. Verify the sign survives midpoint returns, legal delays, and nonoverlapping samples.

CONFUSIONS TO AVOID

Negative autocorrelation is not synonymous with profitable mean reversion. Bid-ask bounce can reverse transaction prices while the executable midpoint offers no edge.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Signal specifications bind the sign claim to a target and executable price. Risk specifications use it to adjust horizon scaling and uncertainty.

RELATED CONCEPTS AND QUICK RECALL

Related: momentum, reversal, bid-ask bounce, horizon aggregation. Recall task: Give economic and mechanical causes of negative lag-one returns. State the price series that distinguishes them.

Persistence is not mean reversion

INTERMEDIATE

ONE-SENTENCE MEANING

Persistence describes how strongly a process remembers its state, while mean reversion describes movement toward a stable conditional center. A persistent process may revert slowly, drift, or contain permanent shocks.

MEMORY JOGGER

Ask what is remembered and what anchor pulls it back. High level autocorrelation alone does not prove a tradable equilibrium.

WHY IT MATTERS IN QUANT RESEARCH

Confusing persistence with reversion creates oversized convergence bets in near-unit-root spreads. Holding period and capital usage depend on the decay mechanism.

FORMULA INTUITION

An AR(1) with $0 < \phi < 1$ reverts but becomes extremely slow as ϕ approaches one. At $\phi=1$, shocks are permanent and the unconditional mean disappears.

SIMPLE MARKET EXAMPLE

A spread with $\phi=0.98$ has half-life near 34 periods. Borrow costs and structural change may arrive before expected convergence.

PYTHON / SYSTEM IMPLEMENTATION

Estimate persistence with intervals and compare random-walk and break-aware challengers. Translate the full range into holding periods and losses.

CONFUSIONS TO AVOID

A high R-squared relation between drifting prices can create persistent residuals without equilibrium. Later stationarity and convergence must support the anchor.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model card records equilibrium, persistence interval, horizon, and invalidation triggers. Limits use the slow plausible decay case.

RELATED CONCEPTS AND QUICK RECALL

Related: unit root, near-unit root, equilibrium, shock decay. Recall task: Explain why $\phi=0.98$ can be stationary but operationally unusable. Name two costs competing with half-life.

Mechanical autocorrelation

INTERMEDIATE

ONE-SENTENCE MEANING

Mechanical autocorrelation is dependence introduced by sampling, smoothing, overlap, staleness, or price conventions. It can be stable and significant while offering no executable forecast.

MEMORY JOGGER

Before explaining a lag pattern, reproduce it from the pipeline alone. A transformation can manufacture memory from independent innovations.

WHY IT MATTERS IN QUANT RESEARCH

Overlapping returns exaggerate evidence, moving averages create persistence, and stale prices delay reactions. These effects contaminate signals and standard errors.

FORMULA INTUITION

An h -period overlapping return shares $h-1$ component returns with its neighbor. A moving-average feature similarly reuses most of its window.

SIMPLE MARKET EXAMPLE

A five-day return computed daily is serially dependent because four daily returns are shared. Each row is not a fresh five-day experiment.

PYTHON / SYSTEM IMPLEMENTATION

Maintain nonoverlapping benchmarks and simulate IID increments through the transform. Compare trade-price and midpoint returns to diagnose bounce.

CONFUSIONS TO AVOID

Deleting one lag does not remove longer overlap. Forward-filling missing prices creates artificial zero returns followed by delayed jumps.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature registry records lookback, stride, gap policy, smoothing kernel, and availability. Validation derives purge rules from this metadata.

RELATED CONCEPTS AND QUICK RECALL

Related: overlapping labels, smoothing, stale quote, nonsynchronous trading. Recall task: Derive the overlap in adjacent five-day returns. State the validation change required by it.

Mean reversion needs a defensible anchor

INTERMEDIATE

ONE-SENTENCE MEANING

Mean reversion is movement toward a defined equilibrium or conditional center. The anchor may evolve, but it must be estimable without future data.

MEMORY JOGGER

A deviation matters only relative to an anchor that survives later observations. Define who owns the center, how it changes, and when it is invalid.

WHY IT MATTERS IN QUANT RESEARCH

Contrarian signals, spread trades, and inventory controls rely on reversion. A weak anchor turns permanent repricing into repeated false entries.

FORMULA INTUITION

Let $e_t = X_{t-m_t}$, with $E[e_{t+1}|e_t]$ closer to zero than e_t . Reversion of the residual does not imply a flat raw level.

SIMPLE MARKET EXAMPLE

A commodity basis may revert around carrying cost rather than zero. Storage, funding, and seasonality move the relevant equilibrium.

PYTHON / SYSTEM IMPLEMENTATION

Specify anchor inputs, fit window, cadence, uncertainty, and event overrides. Backtest frozen historical centers at every forecast origin.

CONFUSIONS TO AVOID

A historical average is not automatically an equilibrium. Frequent crossings can be noise rather than restoring force.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The equilibrium service publishes center, scale, confidence, and effective timestamp. Monitoring checks both residual behavior and anchor drift.

RELATED CONCEPTS AND QUICK RECALL

Related: conditional mean, equilibrium model, deviation, restoring force. Recall task: Propose an anchor for a futures basis. List economic inputs that could move it permanently.

AR(1) as a discrete reversion model

INTERMEDIATE

ONE-SENTENCE MEANING

A stationary AR(1) models the next deviation as a fraction of the current deviation plus a new shock. It is the simplest bridge from autocorrelation to decay.

MEMORY JOGGER

Each period preserves ϕ of the old deviation. The remainder is expected reversion, while noise can still push the path away.

WHY IT MATTERS IN QUANT RESEARCH

AR(1) is a baseline for spreads, rates, and residuals. Its failures reveal changing centers, nonlinearities, or multiple time scales.

FORMULA INTUITION

$X_t = c + \phi X_{t-1} + \epsilon_t$ with $|\phi| < 1$ has mean $\mu = c / (1 - \phi)$. The h -step mean is $\mu + \phi^h (X_t - \mu)$.

SIMPLE MARKET EXAMPLE

A spread at 2.0 with $\mu = 0$ and $\phi = 0.8$ has expected values 1.6, 1.28, and 1.024. Realized values need not follow that smooth path.

PYTHON / SYSTEM IMPLEMENTATION

Fit an intercept inside each training window and save coefficient covariance and residual scale. Compare residual ACF with a no-change baseline.

CONFUSIONS TO AVOID

OLS bias can be material near one in short samples. A fitted ϕ below one does not prove equilibrium or profitable convergence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The forecast artifact exposes center, ϕ , residual distribution, and horizon path. Sizing consumes predictive uncertainty as well as the mean.

RELATED CONCEPTS AND QUICK RECALL

Related: autoregression, conditional forecast, unconditional variance, innovation. Recall task: Compute a three-step forecast for $\phi = 0.8$. Explain why the path can first move farther from the mean.

Half-life and decay speed

INTERMEDIATE

ONE-SENTENCE MEANING

Half-life is the model-implied time for an expected deviation to shrink by half. It translates persistence into an operational holding-period scale.

MEMORY JOGGER

Half-life describes expected decay, not a guaranteed crossing time. Uncertainty becomes highly asymmetric when persistence is near one.

WHY IT MATTERS IN QUANT RESEARCH

Entry horizon, maximum hold, turnover, and cost budget should align with decay speed. A slow edge can be dominated by financing or opportunity cost.

FORMULA INTUITION

For $0 < \phi < 1$, $h_{\text{half}} = \log(0.5) / \log(\phi)$. The result uses the same units as the observation interval.

SIMPLE MARKET EXAMPLE

Phi values 0.80, 0.95, and 0.99 imply about 3.1, 13.5, and 69 periods. Small changes near one create large horizon changes.

PYTHON / SYSTEM IMPLEMENTATION

Transform the coefficient interval into a half-life interval and cap unstable values. Validate actual trade-duration distributions.

CONFUSIONS TO AVOID

Half-life is not average time to cross the mean or close a trade. Negative phi and moving anchors require other interpretations.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The strategy stores observation unit, estimated interval, target duration, maximum hold, and review trigger. Capacity uses the slow plausible case.

RELATED CONCEPTS AND QUICK RECALL

Related: first-passage time, persistence, holding period, parameter uncertainty. Recall task: Calculate half-life at $\phi=0.95$. Explain why an upper confidence bound near one is a capital risk.

Ornstein-Uhlenbeck process

INTERMEDIATE

ONE-SENTENCE MEANING

The Ornstein-Uhlenbeck process uses continuous drift toward a center plus random diffusion. It is useful when horizons are continuous or observations are irregular.

MEMORY JOGGER

The farther the state is from its center, the stronger expected restoring drift becomes. Noise still causes overshoots and delayed convergence.

WHY IT MATTERS IN QUANT RESEARCH

OU models are baselines for spreads and short rates. Continuous-time parameters provide coherent horizon conversion when discretization is correct.

FORMULA INTUITION

$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t$, with half-life $\log(2)/\kappa$. For interval Δ , $\phi = \exp(-\kappa \Delta)$.

SIMPLE MARKET EXAMPLE

If $\kappa = 0.25$ per day, half-life is 2.77 days and one-day ϕ is 0.779. Weekly sampling produces a much smaller discrete coefficient.

PYTHON / SYSTEM IMPLEMENTATION

Estimate with actual time gaps or an appropriate likelihood. Store parameter units, discretization, residual checks, and decision-time mapping.

CONFUSIONS TO AVOID

Continuous-time notation does not remove microstructure noise or jumps. A moving equilibrium and regime shifts violate simple OU assumptions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model layer publishes horizon-specific means and variances. Execution separately applies trading hours, latency, and fill constraints.

RELATED CONCEPTS AND QUICK RECALL

Related: diffusion, restoring drift, continuous time, discretization. Recall task: Convert $\kappa = 0.25$ per day to half-life and daily ϕ . Explain what irregular observations change.

Z-scores and standardized deviations

INTERMEDIATE

ONE-SENTENCE MEANING

A z-score expresses a deviation from an estimated center in estimated scale units. It is comparable only when both estimates are causal and healthy.

MEMORY JOGGER

Z measures unusualness under a reference window. It does not explain the cause or guarantee reversal.

WHY IT MATTERS IN QUANT RESEARCH

Standardized spreads drive entries and cross-sectional ranking. Scale adaptation can control risk or hide a widening structural break.

FORMULA INTUITION

$z_t = (X_t - m_t) / s_t$. A causal version uses m_t and s_t available before the decision, often ending at $t-1$.

SIMPLE MARKET EXAMPLE

A spread at 4 with center 1 and scale 1.5 has $z=2$. If crisis scale doubles, the same raw deviation falls to $z=1$.

PYTHON / SYSTEM IMPLEMENTATION

Version center, scale estimator, lookback, lag, minimum history, and winsorization. Stress denominator collapse during stale data.

CONFUSIONS TO AVOID

A large z-score is not a p-value, and rolling normalization does not ensure stationarity. Including the current extreme shrinks its own score.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature service publishes raw deviation, center, scale, z, confidence, and earliest action. Monitoring separates numerator drift from denominator failure.

RELATED CONCEPTS AND QUICK RECALL

Related: rolling normalization, robust scale, threshold signal, leakage. Recall task: Compute a z-score. Explain how including the current observation in the scale window changes entry timing.

Crossings and first-passage time

INTERMEDIATE

ONE-SENTENCE MEANING

First-passage time is the random time required to reach a boundary such as the mean, exit band, or stop. It describes duration and path risk more directly than half-life.

MEMORY JOGGER

Expected decay asks where the average path goes. First passage asks when a noisy path actually touches a target.

WHY IT MATTERS IN QUANT RESEARCH

Capital lockup, financing, and stop probability depend on passage behavior. Equal half-lives can produce different exits because volatility differs.

FORMULA INTUITION

$\tau_b = \inf\{t > 0: X_t \text{ reaches boundary } b\}$. Its distribution depends on starting distance, drift, diffusion, and monitoring frequency.

SIMPLE MARKET EXAMPLE

A noisy spread can cross zero quickly despite weak restoring force. Frequent crossings therefore do not prove strong mean reversion.

PYTHON / SYSTEM IMPLEMENTATION

Estimate passage and censoring distributions in walk-forward trades. Simulate fitted models with discrete monitoring and execution latency.

CONFUSIONS TO AVOID

Dropping unclosed trades creates survivorship bias. Counting an unexecutable intrabar touch overstates exits.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The simulator records entry distance, first observable touch, first executable fill, censoring, and competing stop. Treasury uses duration quantiles.

RELATED CONCEPTS AND QUICK RECALL

Related: hitting time, survival analysis, censoring, boundary crossing. Recall task: Explain why high noise can shorten crossing time without strengthening equilibrium. Handle trades still open at test end.

Cointegrated spreads

INTERMEDIATE

ONE-SENTENCE MEANING

Cointegration describes a stable combination of nonstationary levels. The hedge relation creates a candidate residual even when component prices wander.

MEMORY JOGGER

Two prices can share a tether, but it must survive later data. Freeze the hedge ratio before judging convergence.

WHY IT MATTERS IN QUANT RESEARCH

Pairs, curve, and basis trades often rely on cointegration. It is more coherent than screening only for high level correlation.

FORMULA INTUITION

If Y_t and X_t are $I(1)$ but $e_t = Y_t - \beta X_t$ is $I(0)$, they are cointegrated. Deterministic terms also define the residual.

SIMPLE MARKET EXAMPLE

Two share classes may track one issuer, yet conversion rules and borrow change the practical anchor. Full-sample beta can hide relation breaks.

PYTHON / SYSTEM IMPLEMENTATION

Estimate beta in each training window and test later residual stability. Simulate actual leg prices, borrow, and asynchronous fills.

CONFUSIONS TO AVOID

Trending levels can correlate without cointegration. Frequent beta refits can manufacture a smooth historical spread.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The spread artifact binds universe, coefficients, fit cutoff, residual, break monitor, and exit policy. Execution receives hedge units.

RELATED CONCEPTS AND QUICK RECALL

Related: error correction, hedge ratio, spurious regression, relative value. Recall task: Explain why high correlation is insufficient for a pairs trade. Identify what must be frozen out of sample.

Error-correction dynamics

INTERMEDIATE

ONE-SENTENCE MEANING

An error-correction model links short-run changes to the prior equilibrium deviation. It separates the long-run tether from the mechanism that repairs imbalance.

MEMORY JOGGER

Levels define the relation while changes perform adjustment. The adjustment coefficient identifies which variable responds.

WHY IT MATTERS IN QUANT RESEARCH

Error correction supports direction and hedge interpretation in cointegrated systems. It can reveal that one leg performs most of the repair.

FORMULA INTUITION

$\Delta Y_t = \alpha(Y_{t-1} - \beta X_{t-1}) + \sum \Gamma_i \Delta Z_{t-i} + \epsilon_t$. Sign interpretation depends on residual normalization.

SIMPLE MARKET EXAMPLE

If futures move while spot is sticky, futures may perform most short-run adjustment. Symmetric leg assumptions then misread the mechanism.

PYTHON / SYSTEM IMPLEMENTATION

Fit the vector and adjustment terms in a leakage-safe sequence. Report impulse paths and uncertainty, not alpha alone.

CONFUSIONS TO AVOID

A significant adjustment coefficient does not ensure profitable convergence. Weak identification, sign conventions, and breaks can overturn the story.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry stores equilibrium vector, adjustment matrix, lag order, normalization, and clocks. Attribution separates long-run and short-run effects.

RELATED CONCEPTS AND QUICK RECALL

Related: VECM, adjustment speed, long-run relation, short-run dynamics. Recall task: Interpret a negative adjustment coefficient under a defined residual. Explain why normalization must be stored.

Nonlinear, asymmetric, and regime reversion

INTERMEDIATE

ONE-SENTENCE MEANING

Reversion speed can depend on deviation size, direction, volatility, or market state. Small moves may drift while large moves either correct faster or signal a broken anchor.

MEMORY JOGGER

Do not force every state into one slope. Ask whether the same market mechanism governs both sides and all regimes.

WHY IT MATTERS IN QUANT RESEARCH

Nonlinear models can align entries with cost bands and stress states. They also multiply parameters and reduce independent episode evidence.

FORMULA INTUITION

A threshold AR uses different ϕ values by band, while a switching model uses state-specific centers and speeds. Causal state probabilities replace retrospective labels.

SIMPLE MARKET EXAMPLE

An ETF discount may drift inside creation costs and close quickly outside them, except during impaired arbitrage. One linear coefficient averages incompatible states.

PYTHON / SYSTEM IMPLEMENTATION

Choose boundaries from economics or nested training search. Validate nearby values, side-specific counts, transition delay, and wrong-state loss.

CONFUSIONS TO AVOID

Optimizing states on final PnL creates severe multiplicity. Smoothed regime labels use future observations and are illegal trading inputs.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The policy versions bands, hysteresis, state probabilities, caps, and fallback. Monitoring tracks occupancy and outcomes by state.

RELATED CONCEPTS AND QUICK RECALL

Related: threshold AR, asymmetry, Markov switching, hysteresis. Recall task: Design a three-band policy with uncertain stress state. Explain how costs determine the inner band.

Window length is a bias-variance decision

INTERMEDIATE

ONE-SENTENCE MEANING

The estimation window determines how much history is pooled under one relationship. Short windows adapt quickly but are noisy, while long windows are precise only if old data remain relevant.

MEMORY JOGGER

Choose a window by later decision loss, not final Sharpe. Strong dependence can make effective evidence far smaller than the row count.

WHY IT MATTERS IN QUANT RESEARCH

Autocorrelation, center, scale, and thresholds all change with window policy. The choice controls responsiveness, turnover, and break delay.

FORMULA INTUITION

A rolling estimator uses $t-L+1$ through t , while an expanding estimator uses all data through t . Exponential weights replace a hard boundary with declared decay.

SIMPLE MARKET EXAMPLE

A 60-day spread model adapts fast but produces unstable ϕ . A 750-day model looks smooth yet can trade a relation broken last year.

PYTHON / SYSTEM IMPLEMENTATION

Compare predeclared rolling, expanding, and decayed challengers at identical origins. Report variance, delay, turnover, and state loss.

CONFUSIONS TO AVOID

Optimizing lookback on the complete backtest consumes the test set. Adjacent rolling estimates look stable because their windows overlap.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry stores lookback, weighting, minimum history, cadence, and warm-up. A window update is treated as a model change.

RELATED CONCEPTS AND QUICK RECALL

Related: forgetting factor, local stationarity, adaptation, effective sample size. Recall task: Compare 60-day and 750-day windows after a break. Name the metrics that decide between them.

Persistence parameter uncertainty

INTERMEDIATE

ONE-SENTENCE MEANING

Persistence estimates are highly uncertain near the unit-root boundary. Small coefficient errors create large changes in half-life and forecast variance.

MEMORY JOGGER

Translate uncertainty into days held, capital tied up, and loss if the anchor fails. Near one, the mapping is strongly nonlinear.

WHY IT MATTERS IN QUANT RESEARCH

Half-life, convergence, stop probability, and leverage depend on persistence. Ignoring uncertainty creates confidence where a random walk is hardest to reject.

FORMULA INTUITION

If ϕ lies in $[0.90, 0.99]$, half-life spans about 6.6 to 69 periods. A symmetric ϕ interval becomes an asymmetric duration interval.

SIMPLE MARKET EXAMPLE

Point $\phi=0.95$ suggests 13.5 periods, while the upper bound implies months rather than weeks. Financing must be sized to the slower case.

PYTHON / SYSTEM IMPLEMENTATION

Use bootstrap, likelihood, or simulation intervals that respect dependence. Propagate draws through forecasts, passage times, and costs.

CONFUSIONS TO AVOID

Clipping ϕ below one to force finite half-life hides risk. Inconclusive unit-root evidence limits confident convergence claims.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The artifact stores coefficient draws or covariance and conservative duration quantiles. Promotion can require intervals inside capital limits.

RELATED CONCEPTS AND QUICK RECALL

Related: local-to-unity, bootstrap, nonlinear transformation, duration risk. Recall task: Transform ϕ interval $[0.90, 0.99]$ to half-life bounds. Choose the bound for capital reservation.

Stationarity before reversion

INTERMEDIATE

ONE-SENTENCE MEANING

A useful reversion model needs a residual with a stable center and decay mechanism over the decision horizon. Tests provide partial evidence, not permanent certification.

MEMORY JOGGER

Test the object the strategy trades with its actual deterministic terms and break assumptions. Then require later convergence evidence.

WHY IT MATTERS IN QUANT RESEARCH

Spurious relations, breaks, and near-unit roots can mimic slow reversion. Misclassification creates repeated entries against permanent repricing.

FORMULA INTUITION

ADF-type tests use a unit-root null, while KPSS-type tests use a stationarity null. Lag and deterministic choices shape both conclusions.

SIMPLE MARKET EXAMPLE

A spread rejects a unit root over ten years but fails after a market redesign. One full-history verdict hides the current failure.

PYTHON / SYSTEM IMPLEMENTATION

Use opposing tests, break-aware challengers, intervals, and rolling-origin loss. Simulate near-unit and broken processes at the actual sample length.

CONFUSIONS TO AVOID

Rejecting a unit root does not prove fast or profitable reversion. Failing to reject does not prove permanent shocks in a short sample.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Validation stores each specification and fit cutoff. Monitoring emphasizes forecast error, center drift, and passage deterioration.

RELATED CONCEPTS AND QUICK RECALL

Related: unit-root test, KPSS, structural break, specification uncertainty. Recall task: Explain how opposing null tests can both be inconclusive. State the operational response to uncertain half-life.

Residual whiteness and Ljung-Box tests

INTERMEDIATE

ONE-SENTENCE MEANING

Residual whiteness means a fitted model removed the linear structure it claims to explain. It does not prove the model or eliminate nonlinear dependence.

MEMORY JOGGER

After fitting, inspect what remains across relevant lags and transforms. A good mean model can leave volatility memory.

WHY IT MATTERS IN QUANT RESEARCH

Residual autocorrelation signals omitted lags, seasonality, wrong dynamics, or defects. It invalidates standard errors that assume independent innovations.

FORMULA INTUITION

The Ljung-Box statistic aggregates squared residual autocorrelations through lag m . Degrees of freedom should reflect fitted parameters.

SIMPLE MARKET EXAMPLE

An AR(1) residual may be clean at lag one yet show weekly settlement structure. A portmanteau test detects the joint pattern.

PYTHON / SYSTEM IMPLEMENTATION

Predeclare lags from horizon and calendar, adjust for parameters, and pair the test with plots. Use robust alternatives under changing variance.

CONFUSIONS TO AVOID

A small p-value does not identify the important lag or economic size. Selecting the cleanest residual among many models creates bias.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The validation report records residual ACF, test results, pair counts, and version. Promotion requires a response to material structure.

RELATED CONCEPTS AND QUICK RECALL

Related: portmanteau test, model residual, omitted dynamics, diagnostic lag. Recall task: Choose Ljung-Box lags for daily data with weekly settlement. Explain why lag choice must be predeclared.

Variance ratio as a horizon diagnostic

INTERMEDIATE

ONE-SENTENCE MEANING

A variance ratio compares multi-period variance with independent-increment scaling. Values below one suggest reversal, while values above one suggest continuation.

MEMORY JOGGER

Under independence, h-period variance grows h times one-period variance. Deviations summarize weighted autocorrelations through the horizon.

WHY IT MATTERS IN QUANT RESEARCH

Variance ratios challenge random-walk and square-root-of-time assumptions. They diagnose a horizon but do not identify a trading mechanism.

FORMULA INTUITION

$VR(h) = \text{Var}(\sum_{i=1}^h r_{t+i}) / (h \text{Var}(r_t))$. It equals one plus a weighted autocorrelation sum under common assumptions.

SIMPLE MARKET EXAMPLE

$VR(5)=0.75$ means five-period changes vary less than IID scaling predicts. True reversal and transaction-price bounce can both explain it.

PYTHON / SYSTEM IMPLEMENTATION

Use heteroskedasticity-robust inference, nonoverlap checks, and predeclared horizons. Compare midpoint and transaction returns.

CONFUSIONS TO AVOID

Overlapping h-period returns induce estimator dependence. Reporting only the lowest ratio among many horizons understates uncertainty.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Risk validation uses VR to challenge time scaling. Signal research still requires executable forecast and cost evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: random walk, horizon scaling, overlap, Lo-MacKinlay test. Recall task: Interpret $VR(5)=0.75$ and give two explanations. State the comparison that distinguishes them.

Hurst exponent and long-memory claims

INTERMEDIATE

ONE-SENTENCE MEANING

The Hurst exponent summarizes scale behavior across horizons under particular models. Its interpretation is fragile under trends, breaks, heavy tails, and short samples.

MEMORY JOGGER

H is a scaling description, not a universal reversion detector. Always state estimator, horizon range, and assumptions.

WHY IT MATTERS IN QUANT RESEARCH

Long-memory claims affect horizon forecasts and strategy narratives. A false anti-persistence label can turn microstructure noise into contrarian alpha.

FORMULA INTUITION

A common relation is $\text{scale}(h)$ proportional to h^H . $H=0.5$ is a benchmark for independent finite-variance increments, not a universal law.

SIMPLE MARKET EXAMPLE

Repeated level shifts can mimic persistence, while tick bounce can produce H below 0.5 at short horizons. Neither defines a stable trade.

PYTHON / SYSTEM IMPLEMENTATION

Compare estimators, scales, detrending, and simulated broken processes. Validate the implied forecast on later data.

CONFUSIONS TO AVOID

H below 0.5 does not guarantee profitable reversion. Log-log plots often look linear over a selected narrow range.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The artifact stores estimator, scale range, preprocessing, uncertainty, and falsification simulations. Production consumes validated forecasts, not H alone.

RELATED CONCEPTS AND QUICK RECALL

Related: long memory, rescaled range, fractional integration, scaling law. Recall task: Explain how a level break mimics long memory. Design one simulation challenging a Hurst result.

Lag and threshold multiplicity

INTERMEDIATE

ONE-SENTENCE MEANING

Multiplicity arises when many lags, windows, pairs, and thresholds are searched. Noise receives repeated chances to look predictive, biasing the selected result.

MEMORY JOGGER

Count the whole research path rather than the final regression. Unreported alternatives still contribute selection pressure.

WHY IT MATTERS IN QUANT RESEARCH

Autocorrelation research creates large search spaces, and trade rules add entries, exits, stops, and holds. Uncontrolled search produces fragile stories.

FORMULA INTUITION

Family-wise false positives rise with the number and dependence of tests. False-discovery control and nested validation address different parts.

SIMPLE MARKET EXAMPLE

Scanning 50 pairs, 20 lags, and 12 thresholds creates 12,000 nominal configurations. The reported top statistic does not carry that uncertainty.

PYTHON / SYSTEM IMPLEMENTATION

Maintain a search ledger, separate discovery from confirmation, and lock a small family before the final holdout. Use economic priors.

CONFUSIONS TO AVOID

A correction cannot repair adaptive preprocessing or repeated manual inspection. A holdout becomes contaminated once it guides redesign.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The tracker records every run, dataset, metric, and choice. Governance limits holdout access and requires fresh forward evidence after redesign.

RELATED CONCEPTS AND QUICK RECALL

Related: multiple testing, data snooping, false discovery, research ledger. Recall task: Compute the nominal search count for 50 pairs, 20 lags, and 12 thresholds. Explain why true search is larger.

Walk-forward validation of reversion

INTERMEDIATE

ONE-SENTENCE MEANING

Walk-forward validation refits anchor, scale, lag model, and thresholds at historical origins. It evaluates only later observations under realistic adaptation.

MEMORY JOGGER

The unit tested is the entire process, not one residual fitted on all history. Every learned quantity belongs inside its fold.

WHY IT MATTERS IN QUANT RESEARCH

Reversion models are vulnerable to full-sample centers, hedge ratios, and regimes. Rolling origins expose unachievable historical stability.

FORMULA INTUITION

At origin t_j , fit A_j through t_j and evaluate on the next block. Purge overlapping outcomes and concatenate only out-of-sample decisions.

SIMPLE MARKET EXAMPLE

A full-sample beta creates a stable spread, but rolling residuals drift and stops cluster after refits. The deployable process is weaker.

PYTHON / SYSTEM IMPLEMENTATION

Predeclare window, cadence, warm-up, search, purge, costs, and fallback. Report by origin, cohort, duration, state, and episode.

CONFUSIONS TO AVOID

Random cross-validation destroys chronology. Tuning fold design after viewing results overfits the validation sequence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The tracker stores one immutable lineage per origin and replays features through fills. Promotion requires reproducible trades, not only a curve.

RELATED CONCEPTS AND QUICK RECALL

Related: rolling origin, purging, fold lineage, temporal generalization. Recall task: List every quantity refit inside a spread fold. Identify which outputs can form final performance.

Point-in-time dependence features

INTERMEDIATE

ONE-SENTENCE MEANING

A point-in-time feature uses only observations, parameters, and transforms available before the decision. This includes bar finalization, revisions, and computation.

MEMORY JOGGER

Move each datum through event, availability, feature, decision, order, and fill clocks. A one-row shift cannot repair an undefined information set.

WHY IT MATTERS IN QUANT RESEARCH

Rolling ACF, z-scores, hedge ratios, and states can leak through current bars or full-history preprocessing. Weak edges are easily dominated by timing leakage.

FORMULA INTUITION

Require event time $\leq$ availability $\leq$ feature cutoff $\leq$ decision $\leq$ order $\leq$ fill. A close-based feature generally cannot fill at that same close.

SIMPLE MARKET EXAMPLE

A daily z-score uses the official close and backtests entry there. The completed signal was not available until after the price.

PYTHON / SYSTEM IMPLEMENTATION

Persist availability, transform state, fit cutoff, version, and earliest action. Add future-mutation tests for feature immutability.

CONFUSIONS TO AVOID

Lagging a final column may ignore timezone, revision, or corporate-action delays. Smoothed regimes remain illegal after shifting.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature platform enforces as-of joins and emits lineage. The simulator consumes earliest-action metadata for legal fills.

RELATED CONCEPTS AND QUICK RECALL

Related: as-of join, information set, bar finalization, feature lineage. Recall task: Audit a close-based z-score through earliest fill. Name timestamps required to prove legality.

Transaction costs define the tradable band

INTERMEDIATE

ONE-SENTENCE MEANING

A forecast is tradable only when expected convergence exceeds spread, fees, impact, financing, borrow, and uncertainty. Costs create a no-trade region around the anchor.

MEMORY JOGGER

The statistical center is not the economic entry threshold. Require conservative net edge before taking risk.

WHY IT MATTERS IN QUANT RESEARCH

Contrarian strategies trade often and provide liquidity when conditions may worsen. Cost error can turn a high hit rate into negative expectancy.

FORMULA INTUITION

Expected net value equals gross convergence minus round-trip costs and risk charge. Thresholds depend on speed, size, volatility, and state.

SIMPLE MARKET EXAMPLE

A two-basis-point reversal cannot support a three-basis-point spread plus impact. Raising the entry band may improve net value while reducing trades.

PYTHON / SYSTEM IMPLEMENTATION

Use size-aware pretrade estimates, fill simulation, adverse states, and postrade calibration. Model entry and exit separately.

CONFUSIONS TO AVOID

A fixed cost hides capacity and stress nonlinearity. Midpoint backtests treat liquidity and adverse selection as free.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Execution publishes cost curves by instrument, size, and state. Policy blocks entries with nonpositive conservative value.

RELATED CONCEPTS AND QUICK RECALL

Related: no-trade region, slippage, market impact, adverse selection. Recall task: Build a net-edge test for a two-leg spread. Explain why exit cost differs from entry.

Signal-to-position mapping

INTERMEDIATE

ONE-SENTENCE MEANING

Signal-to-position mapping converts forecast and uncertainty into bounded exposure. The map should be cost-aware and robust to small estimate changes.

MEMORY JOGGER

A z-score is an input, not a position. Combine expected return, risk, capacity, and model health under explicit constraints.

WHY IT MATTERS IN QUANT RESEARCH

Sizing determines whether model error becomes noise or a portfolio event. Smooth maps reduce chatter but can accumulate risk under a broken anchor.

FORMULA INTUITION

One rule is $\text{position} = \text{clip}(k * \text{expected_net_return} / \text{forecast_variance}, -\text{cap}, \text{cap})$. Dead bands and health discounts are explicit controls.

SIMPLE MARKET EXAMPLE

Two signals have $z=2$, but one predicts ten basis points with stable borrow and the other three with uncertainty. Equal notional is unjustified.

PYTHON / SYSTEM IMPLEMENTATION

Calibrate inside training data, stress parameter error, and test boundary turnover. Separate forecast, risk, and execution inputs.

CONFUSIONS TO AVOID

Linear scaling by z assumes comparable speed and risk. Large deviations may signal structural breaks rather than larger opportunities.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The portfolio engine consumes forecast distribution, cost curve, health, and limits. It logs every component of the target.

RELATED CONCEPTS AND QUICK RECALL

Related: Kelly fraction, risk budget, dead band, exposure cap. Recall task: Design a bounded position rule using net return and variance. State the health condition forcing zero.

Break risk and stop design

INTERMEDIATE

ONE-SENTENCE MEANING

Break risk is the chance that a deviation reflects a changed equilibrium. Stops limit loss under that alternative but must handle noise and gaps.

MEMORY JOGGER

A reversion trade is short a regime change. Define anchor invalidation before seeing the worst historical move.

WHY IT MATTERS IN QUANT RESEARCH

Convergence strategies can win often and lose heavily when relations break. Price, time, model, and portfolio stops address different mechanisms.

FORMULA INTUITION

A price stop limits deviation, a time stop limits duration, and a model stop reacts to instability. Gap size and exit cost determine realized protection.

SIMPLE MARKET EXAMPLE

A spread reaches three sigmas during a scheduled rebalance. A blind price stop exits peak impact, while event controls could reduce exposure earlier.

PYTHON / SYSTEM IMPLEMENTATION

Predeclare stop hierarchy, trigger data, frequency, order type, and re-entry. Stress gaps beyond thresholds and simultaneous liquidity loss.

CONFUSIONS TO AVOID

Optimizing stops on final PnL is data snooping. A stop can truncate recoverable noise yet fail on genuine jumps.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane supports warning, degraded, and hard-stop states. Incidents separate anchor failure, detector delay, and execution loss.

RELATED CONCEPTS AND QUICK RECALL

Related: structural break, time stop, model stop, gap risk. Recall task: Compare price, time, and model stops for a broken spread. Identify which can act before price loss.

Portfolio aggregation of reversion trades

INTERMEDIATE

ONE-SENTENCE MEANING

Portfolio aggregation combines reversion positions while accounting for common anchors, factors, and crowded exits. Pairwise trades can form one concentrated macro bet.

MEMORY JOGGER

Net leg weights are not the whole risk. Map trades to factors, liquidity, venues, counterparties, and failure scenarios.

WHY IT MATTERS IN QUANT RESEARCH

Diversification can vanish when funding stress widens spreads together. Construction must constrain state-dependent covariance and exit demand.

FORMULA INTUITION

Variance is $w^T \Sigma w$, but scenario loss also includes gaps, nonlinear costs, and common stops. Residual-neutral trades can retain large gross exposure.

SIMPLE MARKET EXAMPLE

Ten equity pairs each short expensive growth may look market-neutral. Together they hold one value-versus-growth position.

PYTHON / SYSTEM IMPLEMENTATION

Aggregate physical legs and factors, estimate state covariance, and stress common exits. Allocate by conservative marginal risk.

CONFUSIONS TO AVOID

Zero beta does not imply zero crisis exposure. Pair-level stops do not cap portfolio liquidation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The portfolio layer reconciles trades to positions, factors, liquidity, and scenarios. Reports show common failure contributions.

RELATED CONCEPTS AND QUICK RECALL

Related: factor neutrality, crowding, marginal risk, liquidity scenario. Recall task: Aggregate three neutral pairs into factor exposures. Explain why common stops belong in scenarios.

Production monitoring for reversion models

INTERMEDIATE

ONE-SENTENCE MEANING

Production monitoring tracks data quality, anchor, persistence, duration, costs, and capital sensitivity. A healthy z-score feed can coexist with broken economics.

MEMORY JOGGER

Monitor the input clock, equilibrium, decay, and portfolio consequence. Every alert needs an owner and safe response.

WHY IT MATTERS IN QUANT RESEARCH

Failures arise from stale prices, center drift, slower convergence, borrow loss, or crowded exits. Mechanism diagnostics prevent bad automatic refits.

FORMULA INTUITION

Health combines freshness, residual mean, persistence interval, passage drift, realized cost, stop rate, and target sensitivity. Thresholds need materiality and persistence.

SIMPLE MARKET EXAMPLE

A frozen quote creates zero returns, low scale, and an extreme z-score when trading resumes. Freshness must quarantine the feature first.

PYTHON / SYSTEM IMPLEMENTATION

Implement warning, reduced-risk, and stop tiers with fallback parameters and incident replay. Compare fast and slow estimates.

CONFUSIONS TO AVOID

Automatic refitting can absorb defects or crisis noise. Stable aggregate PnL can hide increasing inventory age.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane binds health to entry permission, caps, refit authorization, and rollback. State changes are logged.

RELATED CONCEPTS AND QUICK RECALL

Related: observability, drift monitor, fallback, incident replay. Recall task: Draft three alert tiers with data, model, and capital triggers. Assign a safe action to each.

Governance and reproducible lineage

INTERMEDIATE

ONE-SENTENCE MEANING

Governance defines ownership, versioning, evidence, and change control for each model and policy. Lineage connects raw observations to historical positions and fills.

MEMORY JOGGER

A spread chart is research evidence, not a production artifact. Deployment includes data, transforms, model, costs, thresholds, and approvals.

WHY IT MATTERS IN QUANT RESEARCH

Small changes to calendars, hedge ratios, windows, or stops can alter behavior materially. Versioning model code alone is insufficient.

FORMULA INTUITION

Artifact identity hashes data snapshot, calendar, transforms, fit cutoff, coefficients, cost model, and action map. Any dependency change creates a version.

SIMPLE MARKET EXAMPLE

A researcher widens entry after a drawdown and overwrites the notebook. Governance requires a new experiment, validation, shadow period, and promotion.

PYTHON / SYSTEM IMPLEMENTATION

Maintain a model card with anchor, assumptions, intervals, break scenarios, monitors, fallback, and retirement. Reproduce representative decisions.

CONFUSIONS TO AVOID

Documentation cannot compensate for weak evidence. Vendor revisions can also defeat replay unless data snapshots are preserved.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry links data, features, forecasts, targets, orders, fills, incidents, and approvals. Deployment rejects unapproved combinations.

RELATED CONCEPTS AND QUICK RECALL

Related: model registry, artifact lineage, shadow deployment, retirement rule. Recall task: List minimum artifacts reproducing one reversion trade. Assign owners to data, model, policy, and execution.

ACF signatures and competing stories

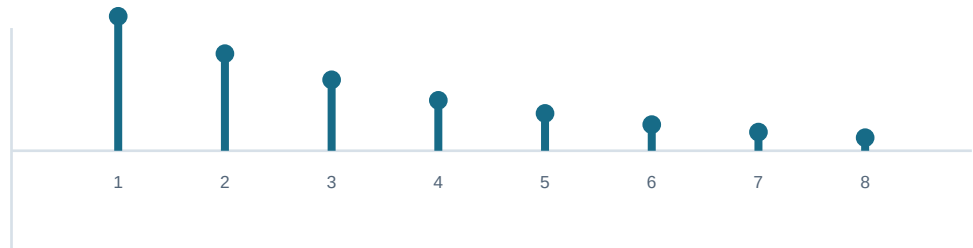
INTERMEDIATE

READING RULE

Treat each profile as a hypothesis generator. The same shape can come from economics, microstructure, a transformation, or mixed regimes, so every story needs a falsification check.

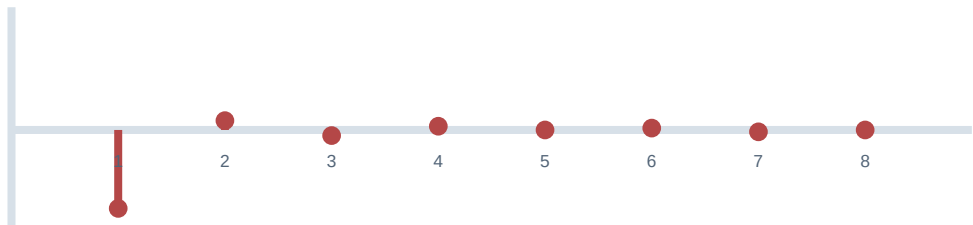
Persistent AR-like decay

Challenge with breaks and slow center drift.



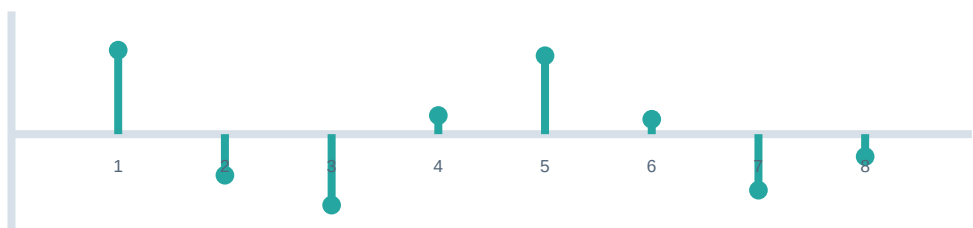
One-lag reversal or bounce

Compare midpoint and transaction returns.



Oscillation or seasonal cycle

Align calendar phase before adding lags.



DECISION IMPLICATION

Choose lags from the decision clock, then validate later forecast loss, residuals, costs, and legal timing. A tall bar is not yet a trade.

The full mean-reversion trade cycle

INTERMEDIATE

CORE IDEA

A successful trade needs more than eventual convergence. The anchor must hold, the signal must precede entry, the path must remain financeable, and the executable exit must preserve net value.

1

Anchor

Economic relation and causal center

2

Deviation

Raw residual, scale, and uncertainty

3

Forecast

Decay path and passage distribution

4

Net edge

Costs, borrow, impact, and risk charge

5

Target

Bounded position and portfolio limits

6

Execution

Legal order, fill, and leg risk

7

Management

Duration, stop, health, and state

8

Exit

Observable boundary and executable fill

9

Learning

Attribution, incident, and approved update

FAILURE ATTRIBUTION

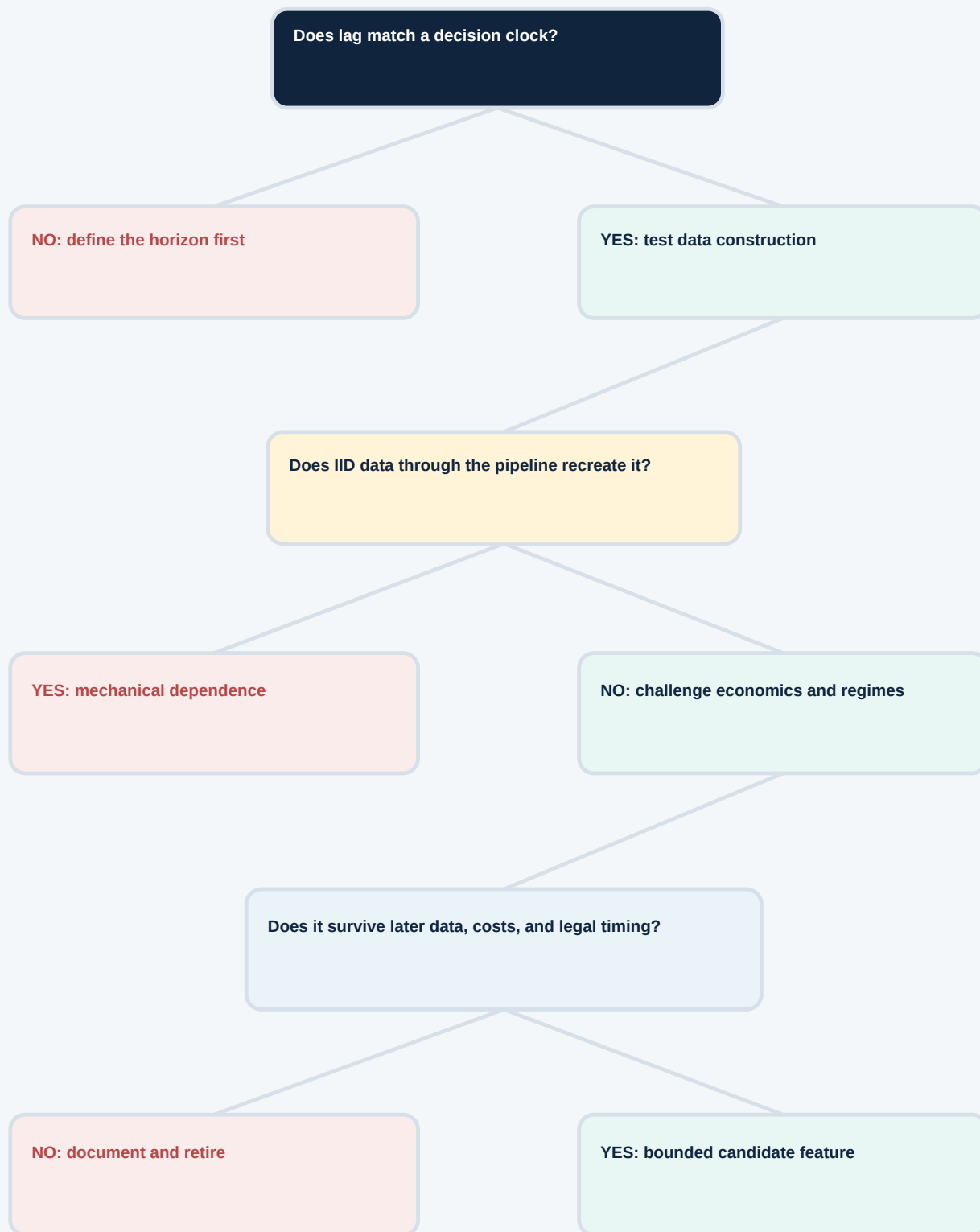
Classify loss as anchor, estimation, timing, cost, execution, state, portfolio, or control failure. The category determines the correct repair.

Choosing and interpreting a lag

INTERMEDIATE

START

Select a lag because it represents a mechanism, horizon, calendar cycle, or diagnostic. If the only reason is that its bar is tallest, independent confirmation is required.



Half-life sensitivity near the unit boundary

INTERMEDIATE

QUESTION

How does a small persistence change alter operational duration across grids? The nonlinear explosion near one is why coefficient uncertainty must be propagated before sizing.

5 min	0.2h	0.3h	0.5h	1.1h	1.9h	0.1d	0.2d
30 min	1.0h	1.6h	0.1d	0.3d	0.5d	0.7d	1.4d
1 hour	1.9h	0.1d	0.3d	0.6d	0.9d	1.4d	2.9d
1 day	1.9d	3.1d	6.6d	13.5d	22.8d	34.3d	69.0d
1 week	9.7d	15.5d	32.9d	67.6d	113.8d	171.5d	344.8d

INTERPRETATION

The same phi means a different calendar duration on a different grid, and phi itself changes under correct aggregation. Compare half-lives only after aligning interval and calendar.

RISK CONSEQUENCE

When a phi interval approaches one, upper duration can become impractical or unbounded. Limits should reflect the slow plausible case.

FORMULA

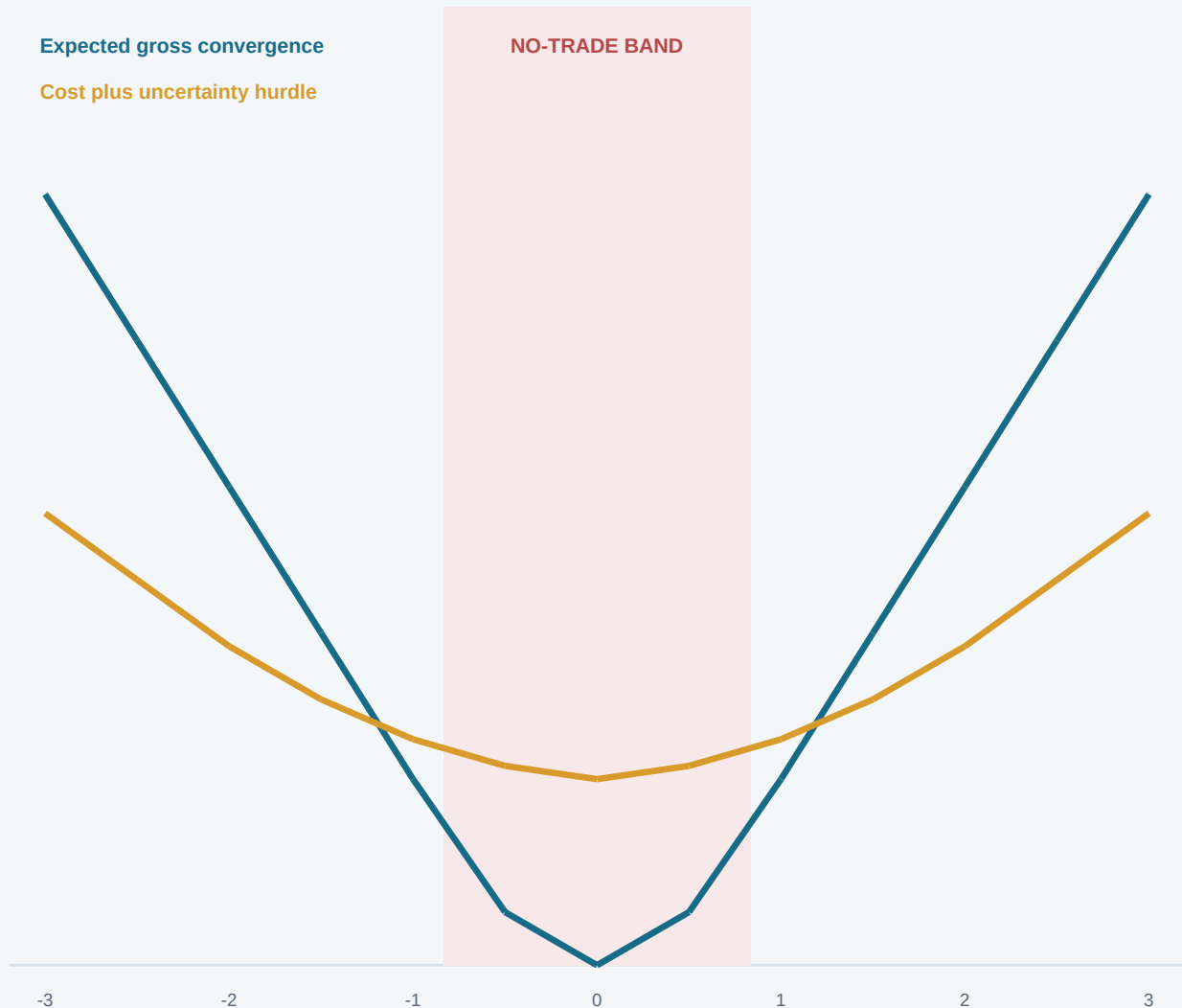
Half-life in periods is $\log(0.5)/\log(\phi)$. Multiply by interval length only when phi was estimated on that same grid.

Deviation, costs, and the no-trade band

INTERMEDIATE

ECONOMIC THRESHOLD

Gross convergence rises with deviation, but costs and break risk can rise too. The deployable band begins where conservative net value is positive, not where z looks unusual.



INTERPRETATION

Where the blue curve is below the gold hurdle, trading destroys expected value. Very large deviations may need extra break-risk penalty, so exposure can flatten rather than grow.

VALIDATION

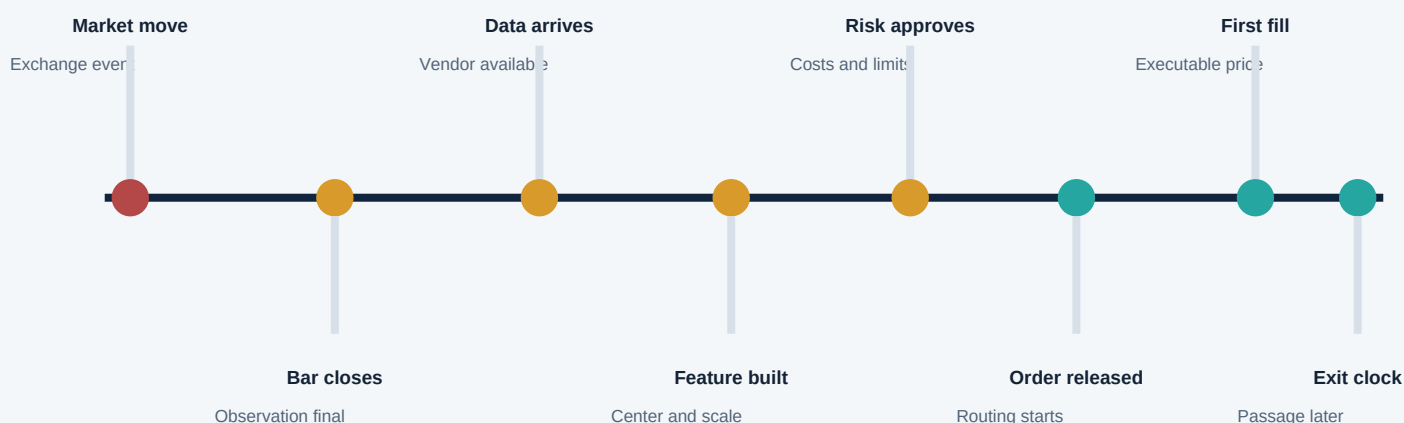
Estimate entry and exit costs by size and state, then perturb both curves. A threshold profitable only under one optimistic cost estimate is not production-ready.

From market event to legal reversion fill

INTERMEDIATE

TIMING STANDARD

Backtests act only after market data, anchor, scale, signal, risk check, and decision exist. Each delay belongs in both trade path and attribution.



TWO COMMON LEAKS

Filling at the close used to calculate the signal assumes impossible simultaneity without an auction process. A smoothed center or regime uses future observations even if the final feature is shifted.

TRADE-DURATION CLOCK

Record first mathematical touch, first observable touch, order release, partial fills, and final exit. Otherwise passage-time research overstates achievable convergence.

AUDIT RECORD

Persist source timestamps, bar rule, feature cutoff, artifact versions, decision time, route, order state, fills, and exit reason. This separates market, data, computation, and execution delay.

Method selection by research question

INTERMEDIATE

QUESTION	FIRST METHOD	KEY ASSUMPTION	PRIMARY RISK
Lag fingerprint	ACF and PACF	Stable local moments	Breaks and multiplicity
Joint residual check	Ljung-Box	Chosen lags	Heteroskedastic size
Horizon scaling	Variance ratio	Comparable increments	Overlap and bounce
Discrete decay	AR model	Stable linear coefficient	Near-unit uncertainty
Irregular-time decay	OU likelihood	Continuous diffusion	Jumps and noise
Stable price relation	Cointegration or VECM	Persistent equilibrium	Hedge-ratio break
Band behavior	Threshold AR	Economic boundary	Tuned thresholds
Changing states	Switching model	Repeated episodes	Label and delay risk

ESCALATION RULE

Use the simplest representation matching the decision and observed failure. Add nonlinear or state complexity only when it fixes a repeatable later-data failure.

Failure modes and diagnostic actions

INTERMEDIATE

SYMPTOM	LIKELY CAUSE	CHECK	ACTION
Negative tick ACF	Bid-ask bounce	Midpoint returns	Do not trade print reversal
Smooth rolling ACF	Window overlap	Nonoverlap estimates	Reduce evidence count
Phi near one	Weak anchor	Interval and break tests	Cap duration and exposure
Z-score explodes	Scale collapse or stale data	Freshness and raw scale	Quarantine feature
Fast zero crossings	High noise	Passage path and costs	Require net edge
One crisis drives result	Episode dependence	Episode resampling	Limit claim and size
Residual weekly pattern	Calendar omission	PACF and session keys	Add predeclared effect
Live fills miss exits	Latency or leg risk	Order and touch ledger	Rebuild execution model

TRIAGE ORDER

Verify identity, clock, calendar, staleness, corporate actions, and leg synchronization before changing a model. Inspect anchor and scale before adding complexity.

Formula sheet and interpretation map

INTERMEDIATE

AUTOCORRELATION

$\rho_k = \text{Cov}(X_t, X_{t-k}) / \text{Var}(X_t)$. State series, grid, demeaning, and sample before interpreting sign.

SAMPLE ACF

$r_k = \sum (x_t - \bar{x})(x_{t-k} - \bar{x}) / \sum (x_t - \bar{x})^2$. Later lags have fewer pairs and more selection risk.

AR(1) FORECAST

$X_t = c + \phi X_{t-1} + \epsilon_t$; $\mu = c / (1 - \phi)$. The h-step mean is $\mu + \phi^h (X_t - \mu)$.

HALF-LIFE

$h_{\text{half}} = \log(0.5) / \log(\phi)$ for $0 < \phi < 1$. Report observation units and transform the parameter interval.

OU PROCESS

$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t$. Half-life is $\log(2)/\kappa$ and ϕ over Δ is $\exp(-\kappa \Delta)$.

Z-SCORE

$z_t = (X_t - m_t) / s_t$. Center and scale must be causal, healthy, and versioned; z is not a p-value.

VARIANCE RATIO

$VR(h) = \text{Var}(\text{sum of } h \text{ returns}) / (h \text{ Var}(\text{one return}))$. Values below one can reflect reversal, overlap, or microstructure.

NET DECISION

Expected net value = expected convergence - cost - impact - financing - risk charge. Positive statistical forecast can remain invalid.

Worked calculation: deviation to governed trade

INTERMEDIATE

1. DEFINE THE RESIDUAL

A legal spread closes at 2.40. Training center is 0.60 and causal scale is 0.90, so deviation is 1.80 and $z=2.00$.

2. ESTIMATE DECAY

Fitted ϕ is 0.85 with interval [0.76,0.94]. Point half-life is $\log(0.5)/\log(0.85)=4.27$ daily periods.

3. CARRY UNCERTAINTY

The interval maps to half-lives near 2.5 and 11.2 days. Capital planning uses the slow case.

4. FORECAST THE PATH

Expected deviations after one, three, and five days are 1.53, 1.11, and 0.80. New shocks can reverse this smooth decay.

5. DEFINE THE EXIT

Approved exit is $z=0.5$, equivalent to 0.45 deviation. Expected gross compression is 1.35 spread units.

6. SUBTRACT IMPLEMENTATION

Entry, exit, impact, borrow, and uncertainty total 0.42 units. Net value is 0.93 before portfolio charge.

7. APPLY CONTROLS

A 12-day time stop matches the slow interval, while a model stop reacts to center drift. Size is capped by correlated exposure.

8. PRESERVE TIMING

The close is known after bar finalization. The backtest acts at the next legal point and records actual leg fills.

9. REGISTER EVIDENCE

Store snapshot, timestamps, center, scale, interval, costs, policy, target, orders, fills, exit reason, and health.

Validation scorecard for a reversion strategy

INTERMEDIATE

ANCHOR EVIDENCE

Economic mechanism, frozen estimation, later residual stability, break scenarios, and invalidation events are documented.

DEPENDENCE EVIDENCE

ACF and PACF survive price conventions, nonoverlap checks, later windows, and appropriate uncertainty.

DECAY EVIDENCE

Parameter intervals imply acceptable duration, and observed passage distributions agree after censoring.

TIMING EVIDENCE

Inputs have availability timestamps, features resist future mutation, and fills follow legal action time.

NET-VALUE EVIDENCE

Gross convergence clears size-aware entry, exit, impact, financing, borrow, and uncertainty charges.

PORTFOLIO EVIDENCE

Physical legs, factors, liquidity, common anchors, crowding, and simultaneous stops remain bounded.

OPERATIONAL EVIDENCE

Health thresholds, owners, degraded mode, hard stop, fallback, rollback, and replay are tested.

PROMOTION GATE

Every required row needs evidence; statistical strength cannot compensate for missing timing or execution.

Final active recall and system index

INTERMEDIATE

OBJECT AND CLOCK

Name residual, price convention, grid, availability, lag horizon, and earliest executable action.

DEPENDENCE MAP

Read ACF and PACF as noisy shapes with pair counts, uncertainty, horizon logic, and mechanical challengers.

ANCHOR CONTRACT

Define center, window, scale, cadence, break conditions, and why the residual should remain stable.

DECAY CONTRACT

Translate parameter intervals into paths, half-life, passage times, capital duration, and wrong-model loss.

TRADING CONTRACT

Specify entry, exit, hysteresis, stop, maximum hold, size map, cost hurdle, borrow, and capacity.

VALIDATION CONTRACT

Walk forward every artifact, purge overlap, count episodes, preserve censoring, and disclose the search family.

PORTFOLIO CONTRACT

Aggregate legs, factors, anchors, venues, liquidity, crowding, and common liquidation scenarios.

PRODUCTION CONTRACT

Monitor freshness, center, scale, persistence, passage, costs, stop rate, sensitivity, and artifact health.

FINAL RECALL

Explain why negative ACF can be untradable, why half-life differs from crossing time, and why full-sample spreads fail live.